CAIE BIOLOGY IGCSE

# Organisation of the Organism

Topic 2 — Complete Study Notes

*Cell Structure • Specialised Cells • Levels of Organisation • Microscopy*

*Content in bold is for Extended students*

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01

# Cell Structure

Animal vs Plant vs Bacterial Cells

## Both Animal & Plant Cells

### Nucleus
Control centre — holds DNA, controls the cell

### Cytoplasm
Jelly-like fluid where reactions happen

### Cell Membrane
Security door — controls what enters/exits

### Mitochondria
Power station — makes energy via respiration

### Ribosomes
Tiny factories that build proteins

## Plant Cells Only (Extra!)

### Cell Wall
Stiff casing made of cellulose — gives structure, stops bursting

### Vacuole
Fluid sac storing water, sugars, salts, and pigments

### Chloroplasts
Capture sunlight → make glucose (photosynthesis)

## Bacterial Cells (Prokaryotes)

### Has:
Cell wall, cell membrane, cytoplasm, ribosomes

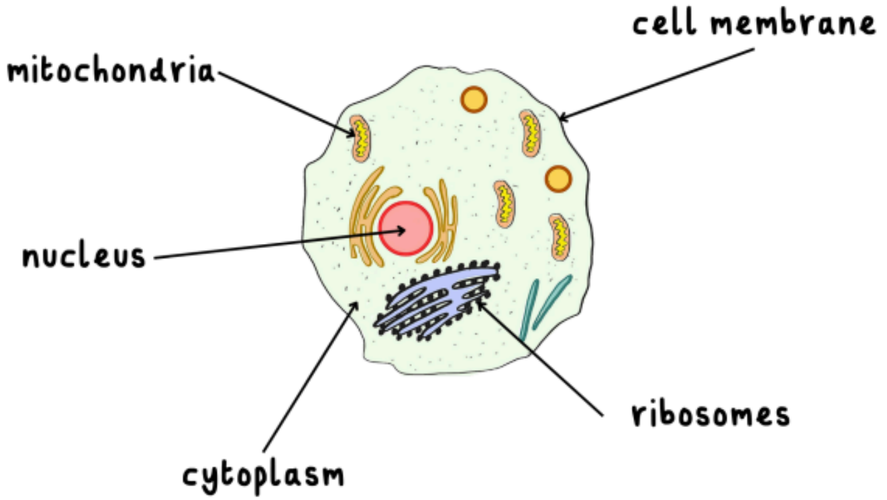
### Cell wall:
Made of peptidoglycan (NOT cellulose)

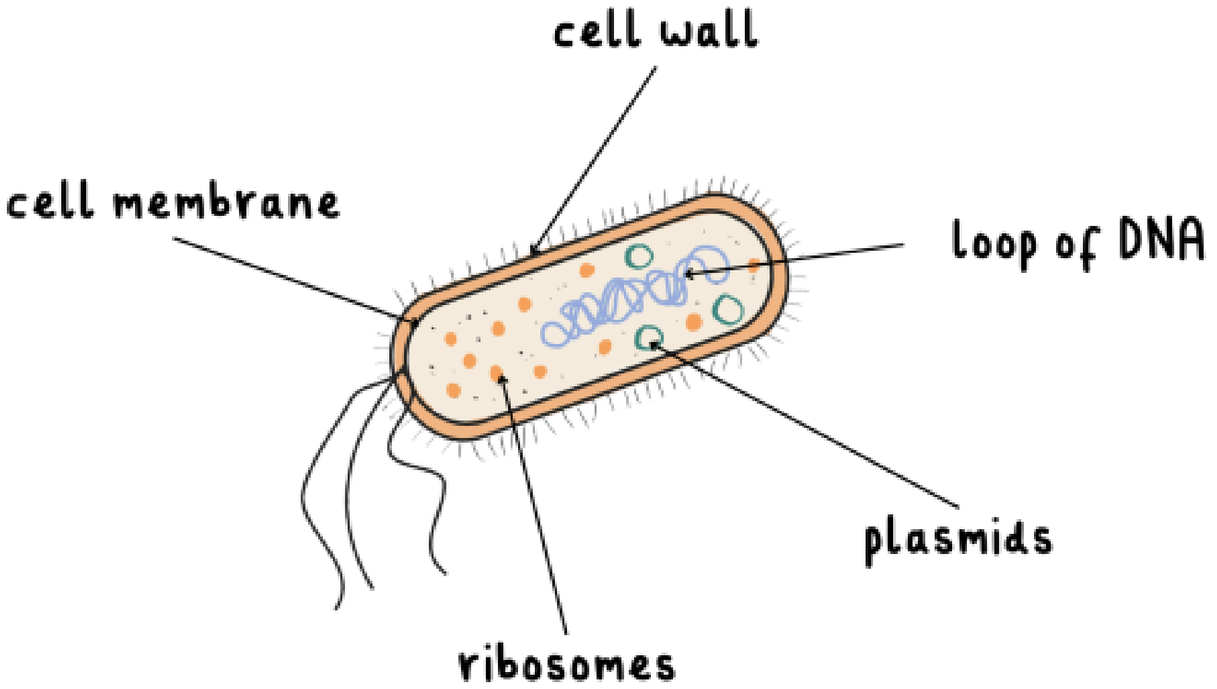
### No nucleus!
DNA floats freely as circular DNA (a loop)

### Plasmids:
Extra tiny rings of DNA

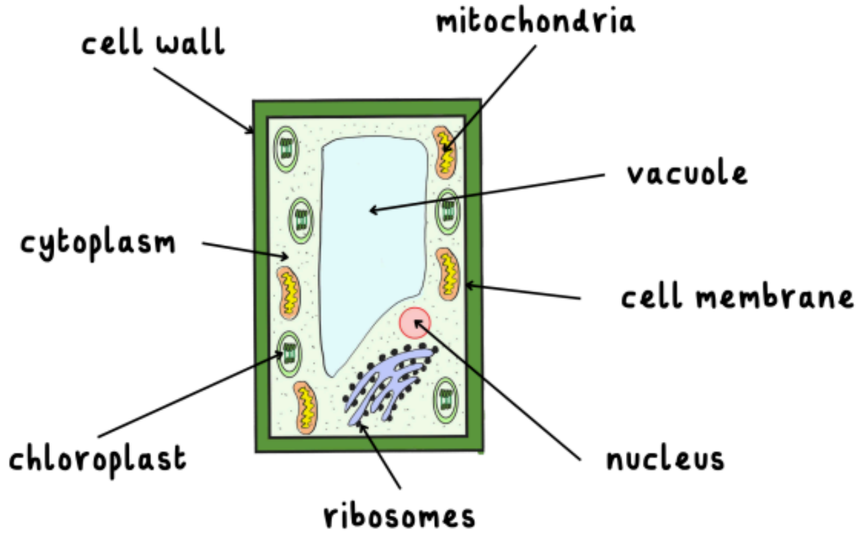
### Missing:
No mitochondria, no chloroplasts

Animal cells




Plant cells



02

# Organelle Functions

What each part of the cell actually does

### Nucleus

Contains DNA (genetic blueprint). Controls all cell activities. Site of DNA replication.

### Cytoplasm

Jelly-like fluid filling the cell. Most chemical reactions of the cell happen here.

### Cell Membrane

Thin, flexible barrier around ALL cells. Controls what enters and exits — like a security guard.

### Mitochondria

Site of aerobic respiration. Converts glucose + oxygen → energy (ATP). More activity = more mitochondria.

### Ribosomes

Smallest organelle. Site of protein synthesis — reads DNA instructions to build proteins.

### Cell Wall

Plants only. Made of cellulose. Gives the cell a rigid shape and prevents it from bursting when full of water.

### Vacuole

Plants only. Large fluid-filled sac. Stores mineral salts, sugars, amino acids, waste, and colour pigments.

### Chloroplasts

Plants only. Site of photosynthesis. Contain chlorophyll (green pigment) to absorb light energy → glucose.

03

# Specialised Cells

*Every cell is shaped perfectly for its job — structure always matches function*

## Red Blood Cells

*Found in: blood*

- Biconcave (dented disc) shape → big surface area
- No nucleus → more space for haemoglobin (carries $O_2$)
- Thin cell membrane → short diffusion distance
- Packed with haemoglobin to carry oxygen

## Ciliated Cells

*Found in: trachea (windpipe) lining*

- Covered in tiny hair-like cilia
- Cilia wave together like a Mexican wave
- Sweep mucus, dust & bacteria upward to throat
- Your body's built-in cleaning system!

## Root Hair Cells

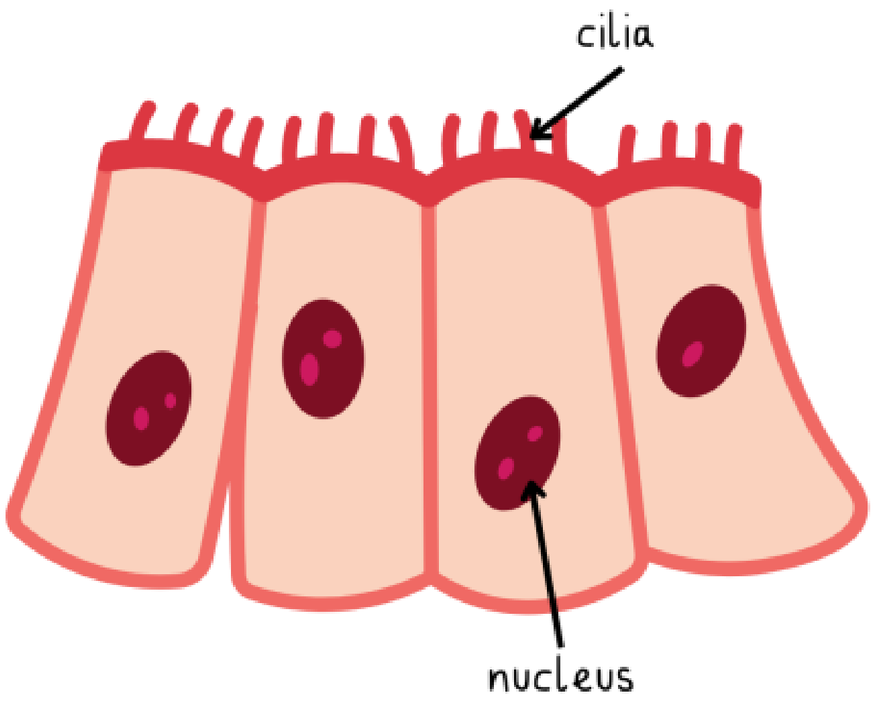
*Found in: plant roots underground*

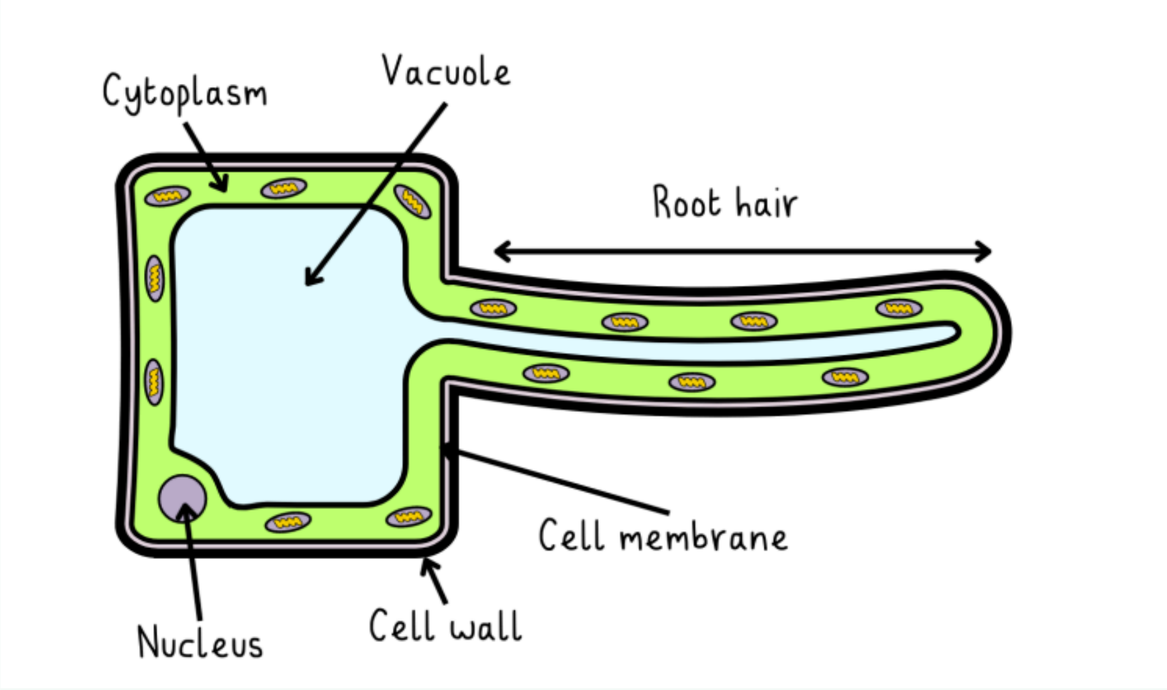
- Long, thin extension into the soil
- Massively increases surface area
- Speeds up osmosis (water absorption)
- Also absorbs mineral ions faster

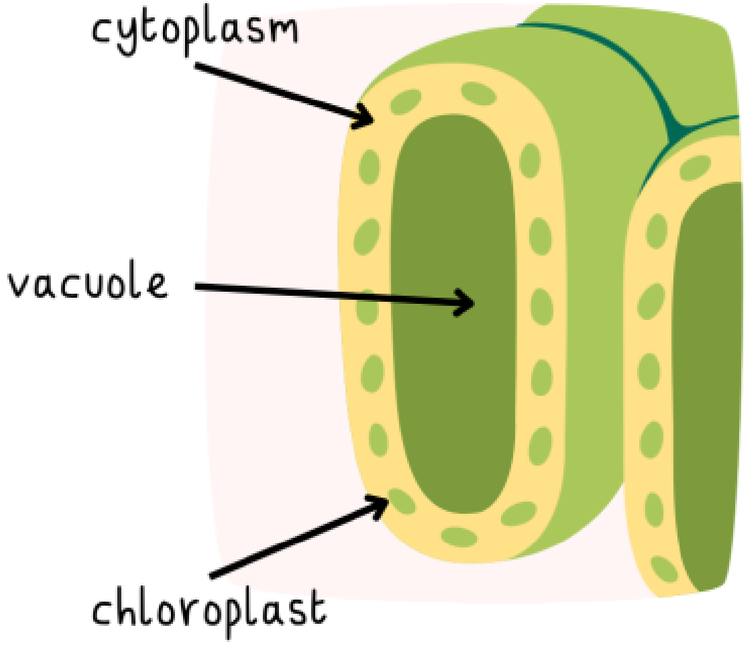
## Palisade Mesophyll

*Found in: top layer of leaves*

- Tall, tightly packed together
- Stuffed with chloroplasts for photosynthesis
- Positioned at top of leaf to catch most sunlight
- Maximum light absorption = maximum food made


cilia
nucleus


Cytoplasm
Vacuole
Root hair
Cell membrane
Nucleus
Cell wall


cytoplasm
vacuole
chloroplast



03

# Specialised Cells (continued)

*Neurones and gametes — extended content*

## ⚡ Neurones (Nerve Cells)

*Found in: brain, spinal cord, and throughout the body*

- **Myelinated (myelin sheath):** Acts like insulation on a wire — stops the electrical signal from slowing down or leaking out
- **Branched dendrites:** Large surface area — can receive signals from MANY other neurons at once
- **Lots of mitochondria:** Nerve cells use huge amounts of energy to fire signals constantly
- **Long axon:** Carries the electrical impulse rapidly over long distances to target muscles or organs

## 🏊 Sperm Cells

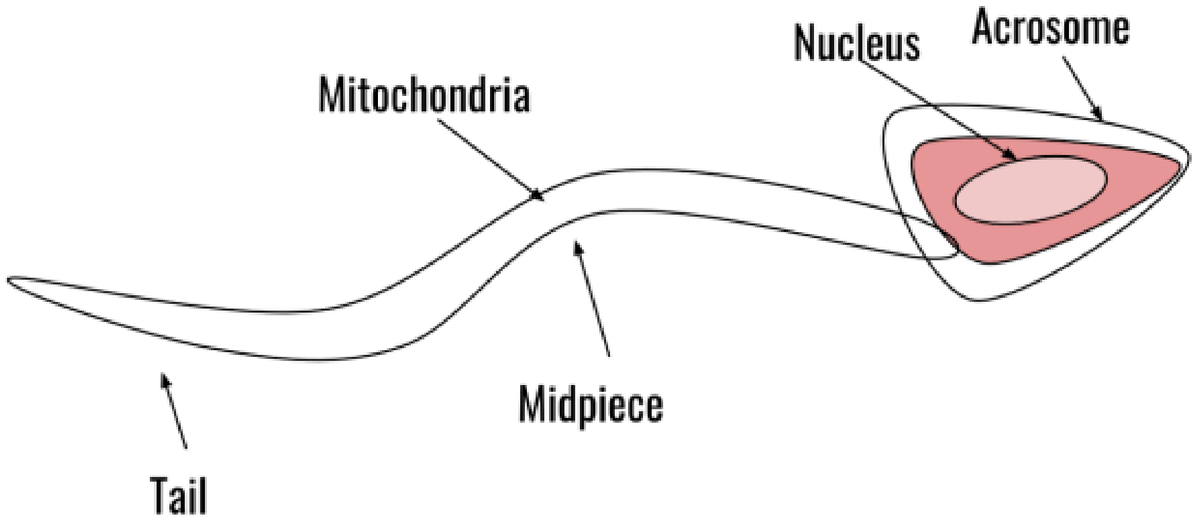
*Found in: male reproductive system (gametes — haploid)*

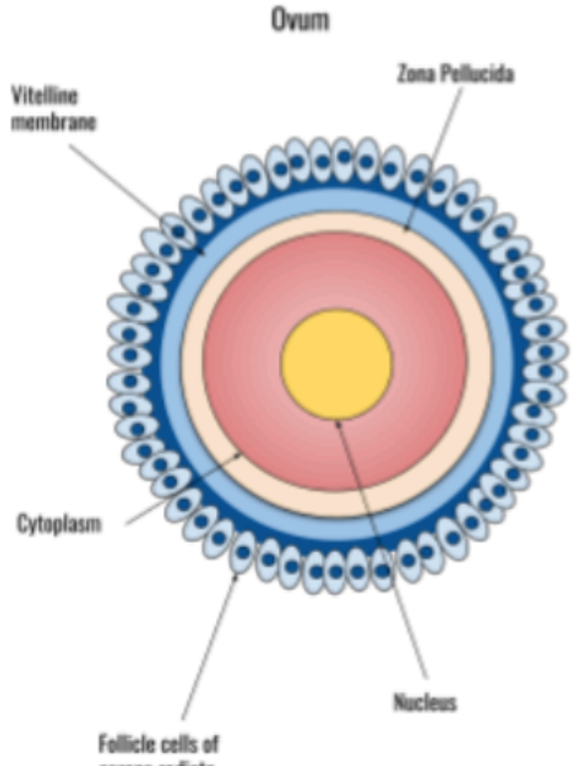
- Tail (flagellum) to swim to egg cell
- Lots of mitochondria in midpiece for swimming energy
- Acrosome on head contains digestive enzymes to penetrate egg membrane
- Haploid — carries only half the chromosomes

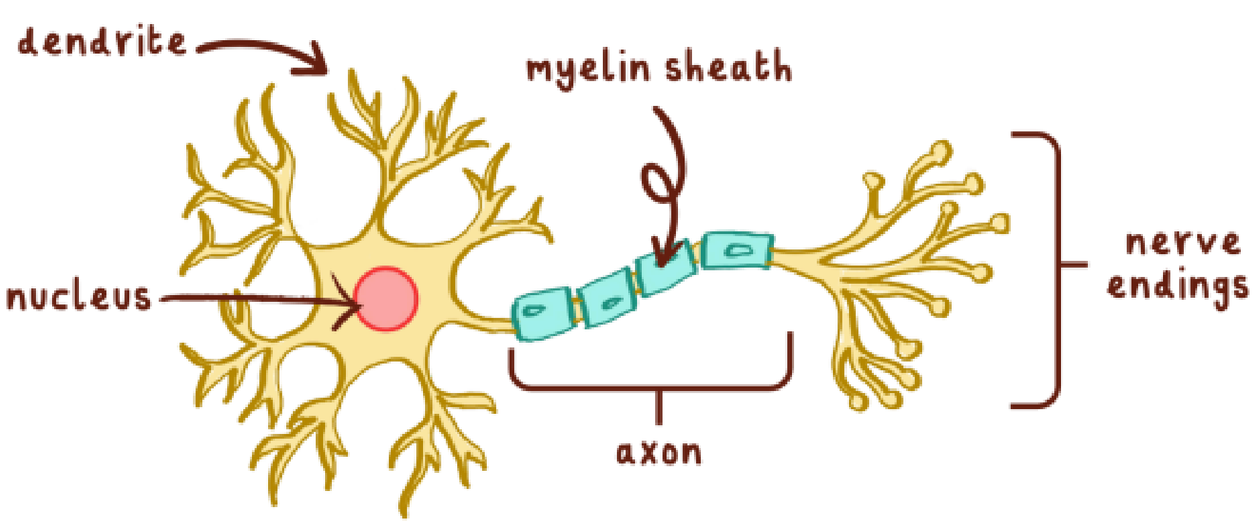
## 🥚 Egg Cells (Ova)

*Found in: female reproductive system (gametes — haploid)*

- Nutrient-rich cytoplasm to feed growing embryo
- Haploid — carries only half the chromosomes
- After fertilisation, membrane hardens to block more sperm
- Largest cell in the human body


Nucleus
Acrosome
Mitochondria
Midpiece
Tail


Ovum
Zona Pellucida
Vitelline membrane
Cytoplasm
Nucleus
Follicle cells of


dendrite
myelin sheath
nucleus
nerve endings
axon

04

# Levels of Organisation

From the smallest unit of life to complete body systems

## CELL 1

The basic building block of all living organisms.
Smallest unit of life that can carry out all life processes.

*e.g. muscle cell, red blood cell*

## TISSUE 2

A group of similar cells working together to carry out a particular process.

*e.g. muscle tissue, blood, epithelial tissue*

## ORGAN 3

A group of different tissues working together to carry out a specific function.

*e.g. heart, lung, stomach, leaf*

## ORGAN SYSTEM 4

A group of related organs working together to carry out major functions in the body.

*e.g. circulatory system, digestive system, respiratory system*

*Key idea: Cells → Tissues → Organs → Organ Systems. Each level adds more organisation and complexity.*

05

# Microscopy Calculations

Calculating actual size from microscope images

THE FORMULA

**Actual Size = Image Size ÷ Magnification**

*Image size is always measured in millimetres (mm)*

**mm → µm conversion**

Multiply by 1000 | 1 mm = 1000 µm | µm = micrometres

**Exam tips**

- Always check if the answer needs to be in mm or µm
- Divide image size by magnification first, THEN convert
- The actual size is always SMALLER than the image size

**Example 1 — answer in mm**

Image size = 32 mm  Magnification = ×100

Actual size = 32 ÷ 100

Answer = 0.32 mm ✓

**Example 2 — answer in µm**

Image size = 54 mm  Magnification = ×500

Step 1: 54 ÷ 500 = 0.108 mm

Step 2: 0.108 × 1000 = 108 µm

Answer = 108 µm ✓

When scientists look at cells under a microscope, the cell appears much bigger than it really is. So we use this formula to figure out the real size:

Actual size = Image size ÷ Magnification

Then if the answer needs to be in micrometres (µm) instead of millimetres (mm), just multiply by 1000.

Example: You see a cell that looks 54 mm wide under a ×500 microscope. 54 ÷ 500 = 0.108 mm → × 1000 = 108 µm — that's the real size.

# Quick Revision Summary

*The 5 things you MUST know for your exam*

**1**

### Cell types

Animal (5 organelles) | Plant (5 + wall, vacuole, chloroplasts) | Bacterial (no nucleus — circular DNA + plasmids)

**2**

### Organelle jobs

Nucleus = control | Mitochondria = energy | Ribosomes = proteins | Membrane = gatekeeper | Chloroplasts = photosynthesis

**3**

### Specialised cells

Structure ALWAYS matches function — every adaptation has a reason. Know WHY, not just WHAT.

**4**

### Organisation levels

Cell → Tissue → Organ → Organ System (think: student → class → school → district)

**5**

### Microscopy formula

Actual size = Image ÷ Magnification. Multiply by 1000 to convert mm → µm.

*Good luck! 🌿 Remember: structure always matches function.*